



Aerodynamic Coefficient Set and 6DOF Simulation

Missile equivalent to the Exocet MM40 Block 2

Team

Gabriel Henrique Alves Rezende
João Pedro Chamhum Basílio
Mateus Pereira Alves
Matheus Marinheiro Spontam
Sávio Lima Rodrigues

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1 Adopted Data

The geometric data and the internal layout of the Exocet MM40 were taken from: <http://sistemasdearmas.com.br/asv/exocet1historia.html>.



Figure 1: External reference used to define the equivalent geometry.

Table 1: Main adopted parameters.

Parameter	Value
Total length	5.79 m
Diameter	0.35 m
Wingspan	1.13 m
Launch mass	875 kg
DATCOM moment reference	2.70 m
Mach range	0.6 to 1.0
Angle-of-attack range	-20 to 20 deg
Control deflection range	-20 to 20 deg

Modeling assumptions

The missile was treated as an equivalent Exocet-class configuration, not as an exact reverse-engineered model. The external dimensions came from the reference page, while the internal mass split was estimated from the subsystem figure and from the previous solid-propulsion sizing. The DATCOM model used an axisymmetric body with two fin sets and a fixed moment reference at $X = 2.70$ m. This point is close to, but not identical to, the estimated initial CG. The 6DOF simulation kept the course Simulink model and replaced only the missile data, CG/inertia data, launch condition, and aerodynamic matrix. The 6DOF run was used as an initial dynamic check of the generated coefficient bank, with emphasis on the alpha versus delta response.



2 Coefficient Generation

In C:\01-Aerodinamica\RunDatcom2, the file `for005.dat` was generated with the equivalent Exocet geometry. The workflow was:

```
for005.dat -> Misdat/RunDatcom2 -> coeficientes.dat -> Coef2Mat -> M_aed.mat
```

The original `RunDatcom2.exe` from the course package was tied to the geometry of the example missile. Running it without changing its source would regenerate coefficients for the original classroom missile, not for the Exocet-equivalent geometry. Therefore, an equivalent MATLAB script was used to write the Exocet `for005.dat`, run `Misdat.exe`, and assemble the same `coeficientes.dat` structure expected by the provided `Coef2Mat.m` script. Thus, the DATCOM processing chain was preserved, but the geometry generation step was adapted to the present missile.

DATCOM setup

The first step was to define the reference center and the control-deflection sweep. The same XCG was later used in the 6DOF model.

```
xcg = 2.700;
delta = -20:2:20;

write_exocet_for005(0, xcg);
```

The geometry file was then regenerated for each deflection. For each case, `Misdat.exe` produced `for006.dat`; the MATLAB reader extracted the coefficients and appended them to `coeficientes.dat`.

```
for k = 1:numel(delta)
    write_exocet_for005(delta(k), xcg);
    status = system('Misdat.exe');

    [mach, alpha, CN, CM, CA, CY, CLN, CLL, CL, CD, XCP, ...] = ...
        DATCOMreader('for006.dat');
end

Coef2Mat('coeficientes.dat');
```

Table 2: Main DATCOM geometry cards used in `for005.dat`.

DATCOM parameter	Value
Moment reference, XCG	2.700 m
Nose type	Ogive
Nose length, $LNOSE$	0.700 m
Body diameter, $DNOSE$	0.350 m
Center-body length, $LCENTR$	5.090 m
Fin set 1 span	0.175 to 0.370 m
Fin set 1 chord	0.450 to 0.200 m
Fin set 1 leading edge, XLE	0.750 to 0.950 m
Fin set 2 span	0.175 to 0.565 m
Fin set 2 chord	1.150 to 0.350 m
Fin set 2 leading edge, XLE	3.800 to 4.650 m
Hinge position	0.975 m



Generated coefficients

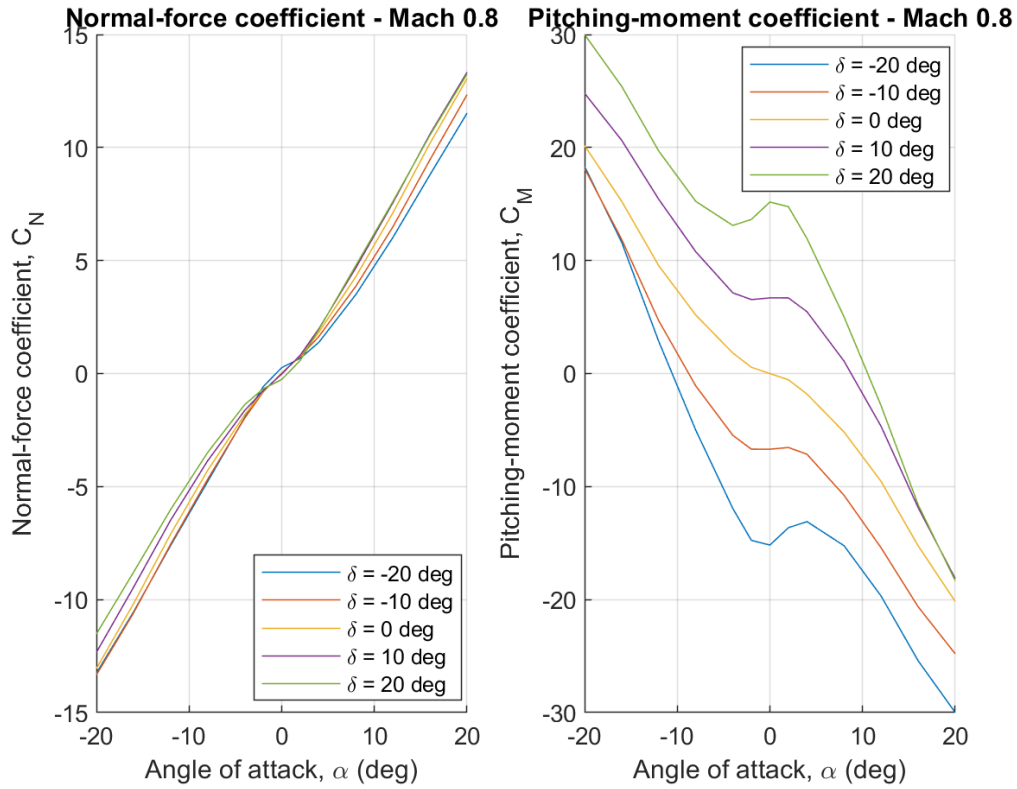


Figure 2: Normal-force and pitching-moment coefficients over the simulated Mach range.

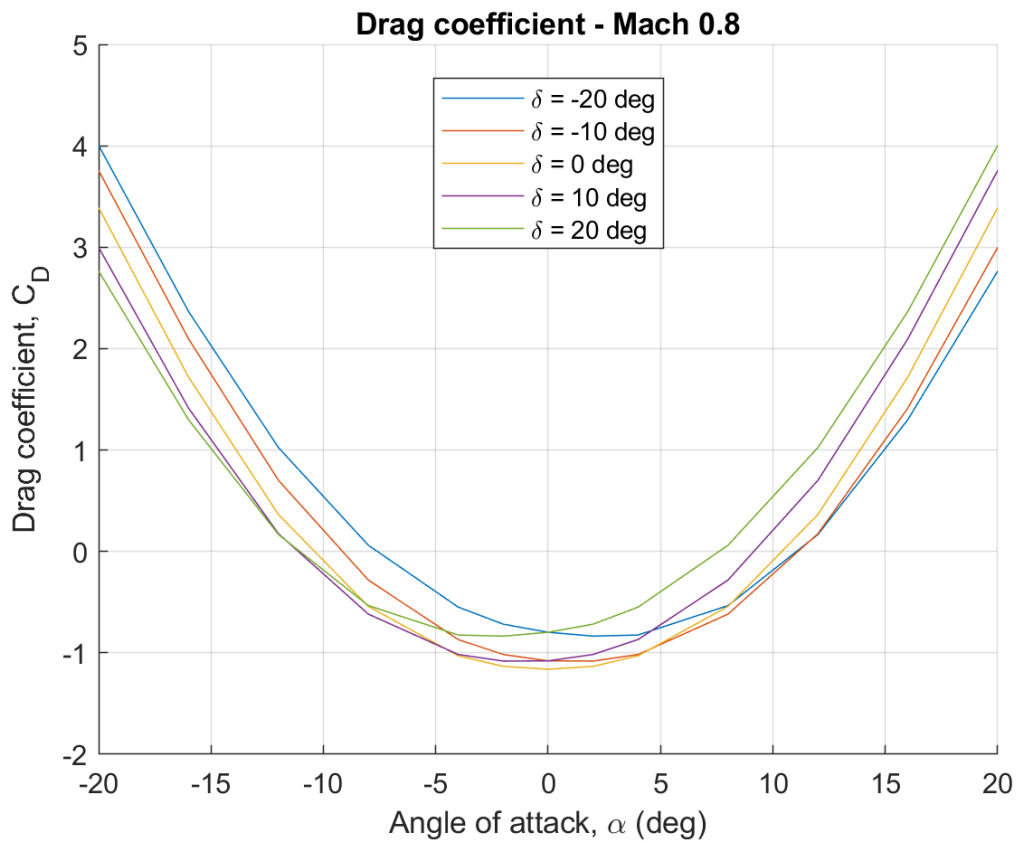


Figure 3: Drag coefficient obtained from the DATCOM coefficient set.



Static stability indication

At Mach 0.9, $\alpha = 0^\circ$, and $\delta = 0^\circ$, the generated table gives $C_{M\alpha} = -15.36 \text{ rad}^{-1}$ and $C_{N\alpha} = 22.70 \text{ rad}^{-1}$. The negative $C_{M\alpha}$ indicates a restoring pitching moment around the DATCOM moment reference. The DATCOM output also reports $X_{CP} = -0.677$, which is dimensionless and given in calibers with respect to the moment reference. Therefore, using $X_{ref} = 2.700 \text{ m}$ and $D_{ref} = 0.350 \text{ m}$,

$$X_{CP,abs} = X_{ref} - X_{CP}D_{ref} = 2.700 - (-0.677)(0.350) = 2.937 \text{ m}.$$

Thus, for the initial estimated CG at 2.833 m, the pressure center is about 0.104 m, or 0.30 calibers, aft of the CG. This is consistent with the negative $C_{M\alpha}$. Transferring the pitching-moment slope to the estimated CG gives $C_{M\alpha,CG} \approx -15.36 + 22.70(0.133/0.350) = -6.73 \text{ rad}^{-1}$, which remains statically stable for this reference condition.

3 CG and Inertia

The CG and inertia estimate was made by sections, using the internal subsystem figure as a guide. The longitudinal axis was measured from the nose. The DATCOM moment reference was also applied in `DadosCGInercia.m`:

$$D.CRM = [2.700 \ 0 \ 0]^T.$$

% Same XCG used in DATCOM.
D.CRM = [2.700 0 0]';

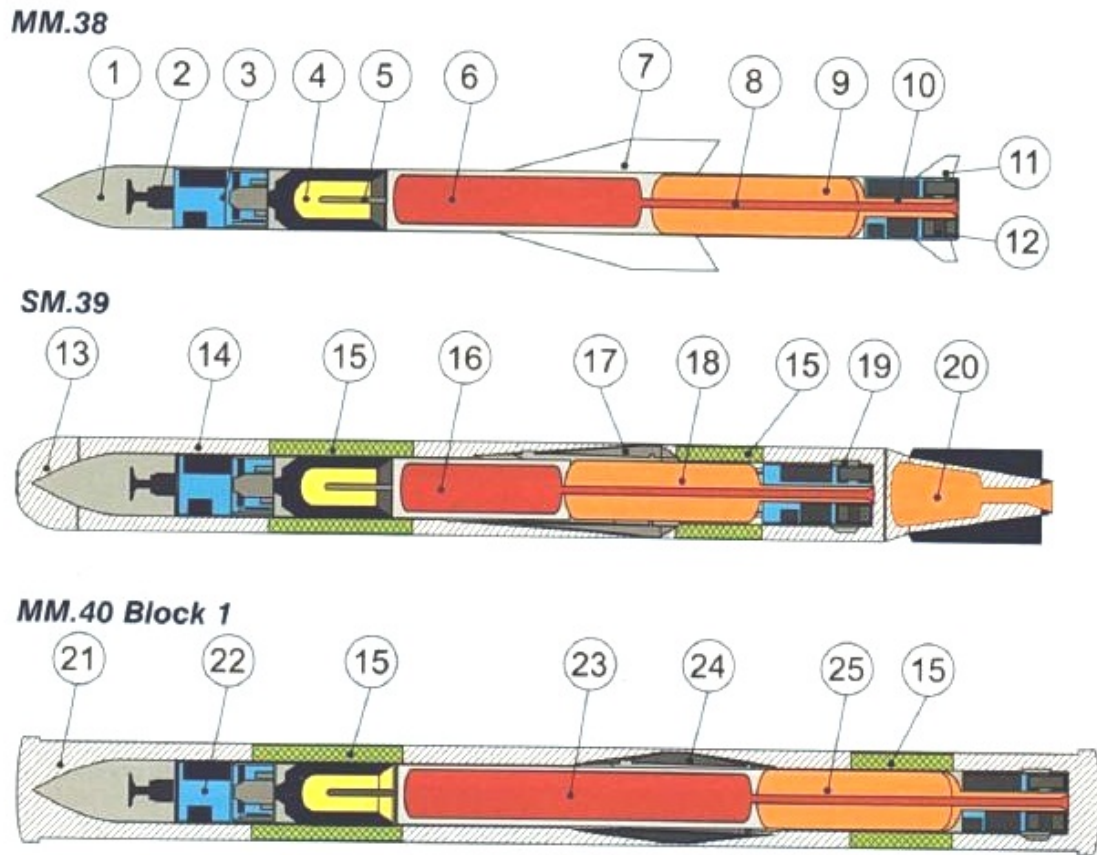


Figure 4: Internal reference used to distribute mass by subsystem.

Table 3: Mass distribution used in the 6DOF model.

Section	Length (m)	Structural mass (kg)	Propellant (kg)
Control/seeker section	0.70	70.0	–
Warhead	0.95	165.0	–
Booster	0.80	45.0	126.4
Sustainer	3.14	70.0	383.5
Nozzle	0.20	15.0	–

With this distribution, the initial mass was 874.9 kg and the total length was 5.79 m. The calculated initial CG was 2.833 m, and the final CG after propellant burnout was 1.856 m.



The 2.700 m value is therefore a common moment reference for the aerodynamic and 6DOF models, not the exact initial CG from the mass breakdown. The difference between the estimated initial CG and this reference is 0.133 m, or 0.38 calibers. Keeping one common moment reference avoids mixing aerodynamic and dynamic reference points, while the mass model itself still carries the time-varying CG used by the 6DOF simulation.

Table 4: Initial and final inertia estimates used in the 6DOF model.

Parameter	Initial value	Final value
Mass	874.9 kg	365.0 kg
X_{CG}	2.833 m	1.856 m
I_{xx}	13.40 kg.m ²	5.59 kg.m ²
$I_{yy} = I_{zz}$	2155.40 kg.m ²	863.30 kg.m ²

The final CG value corresponds to complete propellant burnout. In the short 2.8 s 6DOF test, only the initial booster phase is represented; in that interval, the simulated CG moves from 2.833 m to 2.938 m because the early consumed propellant is located forward of the initial overall CG.



4 6DOF Simulation

In C:\02-Dinamica6DOF, the missile data, CG/inertia, and launch-condition files were edited. The script VooUnico.m was also created. The Simulink model Dinamica6DOF.slx was kept as provided. The M_aed.mat matrix generated from DATCOM was copied to this folder before the simulation.

The test was run during the initial acceleration segment, with an initial speed of 10 m/s and a control deflection of $\Delta Y = 2^\circ$. This low initial speed is a numerical test condition, not a claim about the actual launch speed. It lets the model start smoothly from the beginning of the booster phase and keeps the run focused on the consistency between the generated aerodynamic table and the 6DOF model. Aerodynamic forces in the first instants are therefore lower than they would be in an operational post-launch condition. The final time was limited to 2.8 s to keep Mach inside the main range of the aerodynamic table. At the end of the simulation, the missile reached Mach 0.876 and 294.9 m/s.

```
C.Tfim = 2.8;
C.DeltaY = 2*D2R;
C.DeltaZ = 0*D2R;
C.Vmx0 = 10;
```

6DOF results

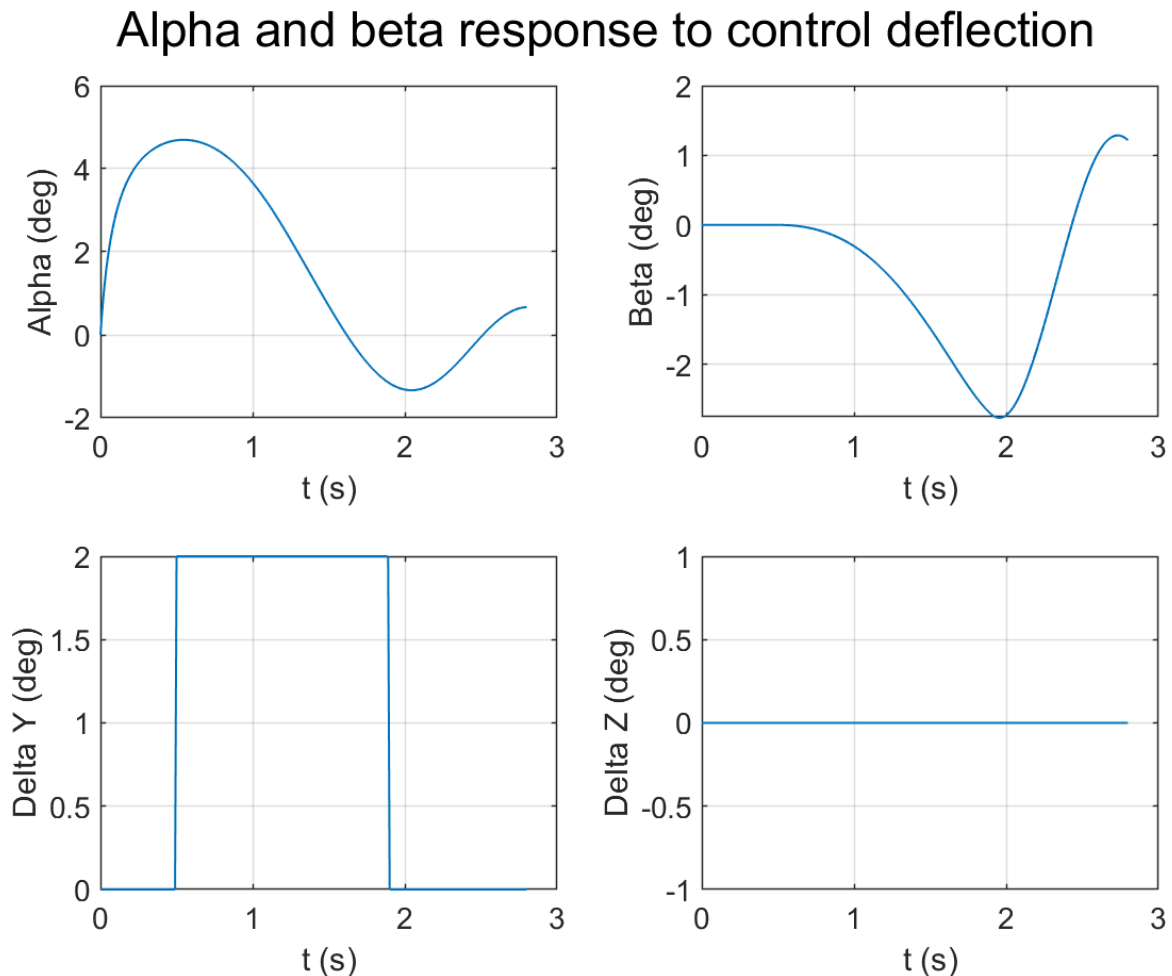


Figure 5: Alpha and beta response to the applied control deflection.

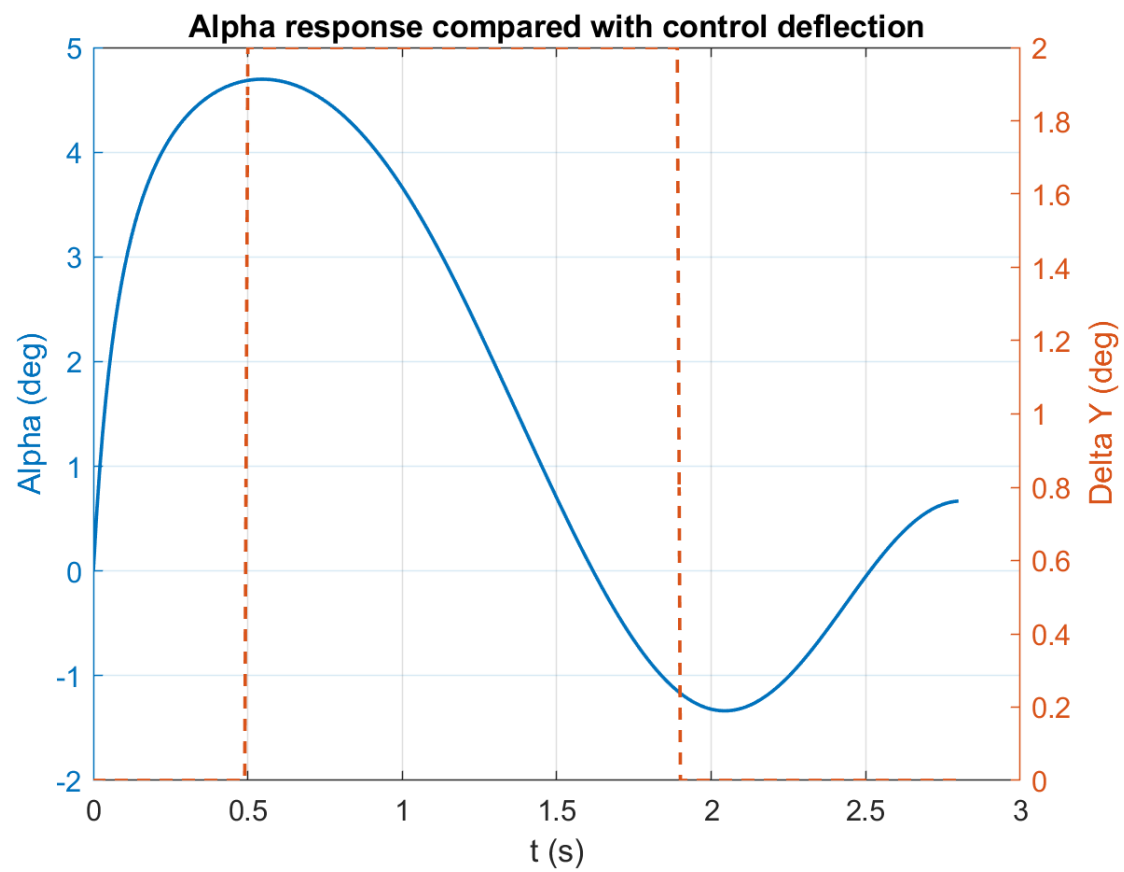


Figure 6: Direct comparison between $\alpha(t)$ and the applied $\Delta Y(t)$.



Velocity and accelerations

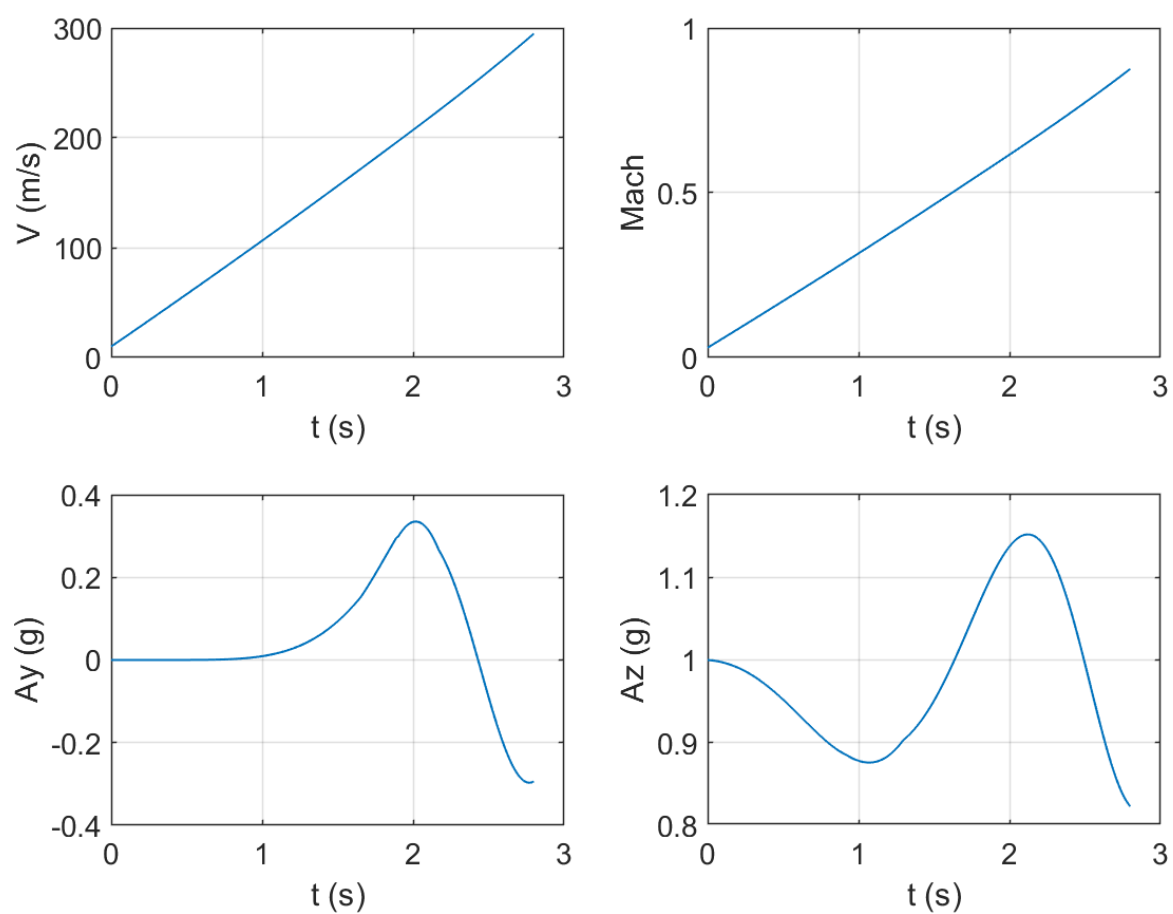


Figure 7: Velocity, Mach number, and accelerations in the 6DOF test.



Propulsion, mass, and inertia

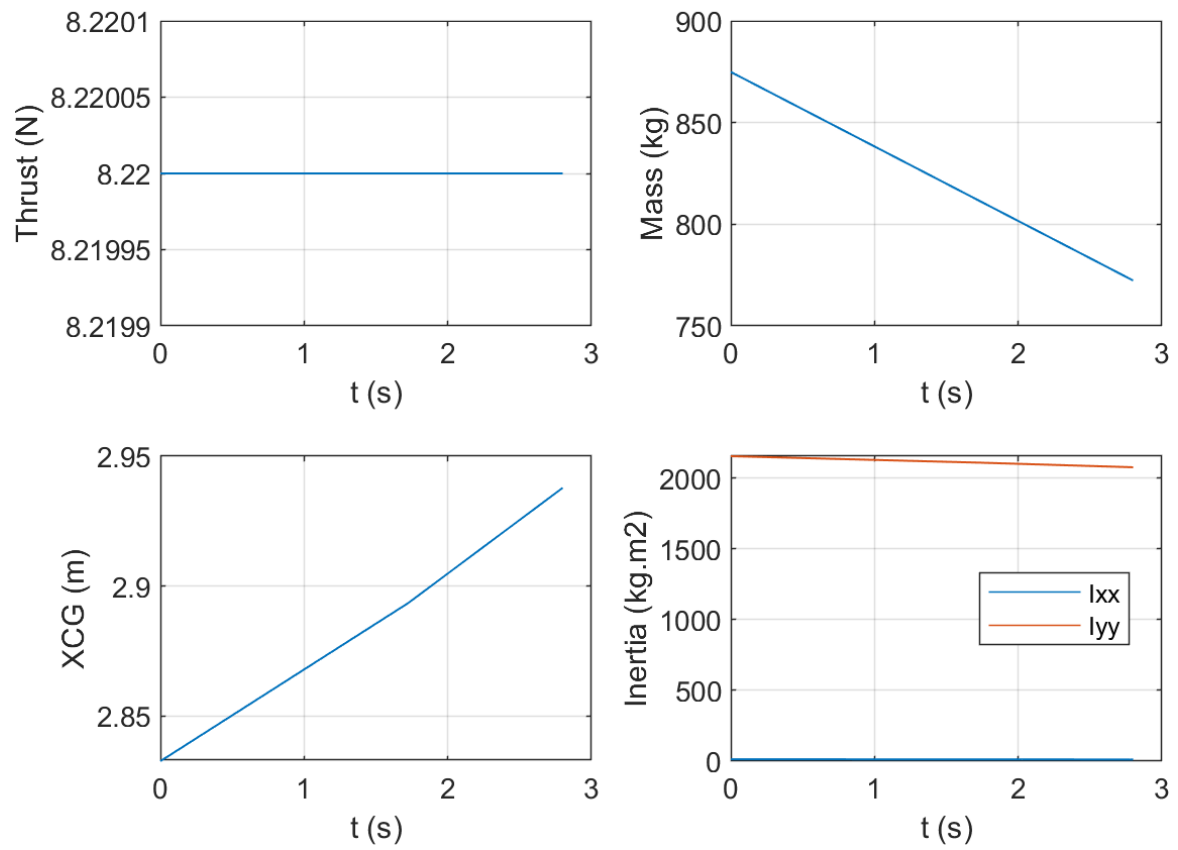


Figure 8: Thrust, mass, CG, and inertia during the initial burn.



5 Result Comments

The aerodynamic coefficient set shows that drag increases with Mach number and angle of attack, as expected for a finned slender body. The pitching moment changes significantly with alpha, and the reference condition at Mach 0.9 gives $C_{M\alpha} < 0$, indicating a restoring pitching trend around the selected moment reference.

In the alpha versus delta result, the 2° deflection produces a limited alpha response, with a peak near 4.7° . The approximate peak ratio is therefore $\alpha_{peak}/\delta \approx 2.35$. Physically, the control surface generates enough moment to perturb the attitude, while the response remains inside the tabulated aerodynamic range. Beta remains smaller than alpha because the command mainly excites the longitudinal plane; the nonzero beta is attributed to the coupled 6DOF equations and remains bounded during the short test.

The missile mass decreases during the booster burn, and the CG moves as propellant is consumed. In the short 2.8 s test the CG moves slightly aft, from 2.833 m to 2.938 m, because the consumed booster propellant is ahead of the initial overall CG. For complete burnout, the estimated CG would move forward to 1.856 m after the sustainer propellant is also consumed. This CG variation changes the moment arms and therefore the dynamic response, which is why the common moment reference and the time-varying CG table must be kept consistent.